Mortgage QC Platform -- AI-Powered Quality Control

The Problem

Mortgage lenders perform QC audits on closed and pre-funding loans. Today this is **sampled at 5-10%** due to manpower constraints. Each audit involves **100-800+ questions** across 17 categories, checking documents against each other and against regulatory requirements. A single loan review takes **4-8 hours**. The process is slow, inconsistent, and incomplete -- deficiencies in unreviewed loans create buyback risk, insurance payouts, and regulatory penalties.

The Solution

This platform uses **AI agents** to automate 100% of QC audits. Each loan submitted for review triggers a workflow that coordinates across **9 lending systems**, running **845 real QC questions** from the lender's questionnaire in parallel, and producing a comprehensive audit report in minutes.

Key Insight: Structured Checks, Intelligent Analysis

The QC questions are known and well-defined. What requires intelligence is cross-referencing data across disconnected systems, identifying discrepancies, applying judgment on tolerances, and adapting to loan type (Conventional vs. FHA vs. VA -- each has different rules). This maps naturally to an agentic architecture where the **workflow is structured but each check is intelligent**.

Architecture

Routes (QC Workflows)

A **route** defines the complete QC workflow for a specific loan type. Routes are DAG-based workflows that fan out parallel checks, merge findings, and generate a final report. All routes are **declarative YAML definitions**.

Route	Loan Type	Blocks
conventional-purchase-pc	Conventional Purchase (Fannie/Freddie)	16 parallel check blocks + report

Route	Loan Type	Blocks
fha-purchase-pc	FHA Purchase	17 blocks (adds FHA compliance)
va-purchase-pc	VA Purchase	17 blocks (adds VA eligibility)

Each route follows the same DAG pattern:

```
intake -> fan_out -> [16-17 check blocks in parallel] -> fan_in merge -> qc_report_gene
```

Blocks (Check Modules)

A **block** is a reusable check module containing a complete question catalog from the lender's QC questionnaire. The platform has **20 blocks** containing **845 real question codes** with **5,389 response variants**, all extracted from actual lender Excel questionnaires.

Block	Questions	Category
Property / Appraisal Review	148	Appraisal completeness, comparables, CU score
Product Specific Check	119	Product eligibility, investor overlays, ARM, renovation
Income Verification	111	W-2, paystubs, tax returns, VVOE, self-employment
Underwriting Review	102	AUS findings, conditions, risk assessment, compensating factors
Credit / Liabilities Review	84	Tradelines, DTI, undisclosed debts, student loans
Asset Verification	75	Bank statements, gifts, earnest money, reserves
Closing Documents Review	34	Note, deed, CD, escrow, e-sign, rescission
Loan Documents Review	33	Purchase contract, title, survey, third-party docs
Appraisal Form 1033	31	Fannie Mae 39-item desktop appraisal review
Insurance Review	28	Hazard, flood, MI, loss payee, condo master policy
Data Validation Services	20	DU validation, AVM comparison, reps & warrants
Application Verification	19	URLA completeness, CIP, LEP, borrower data
Information Integrity	16	Reverification, document age, UDAAP (confirm authority)
Certification / Delivery	10	Endorsement, delivery, investor data compliance

Block	Questions	Category
EPD Review	9	Early Payment Default, fraud indicators (confirm authority)
Compliance Review (ATR/QM)	6	ATR/QM, TRID, HPML, points & fees

Per-Question Haiku Execution

Each check block uses a **per-question execution strategy** controlled by the YAML configuration:

```
step_template:
  agent:
    model: haiku
    per_question: true
```

When `per_question: true` is set, the engine:

- Gathers loan data once (shared tool calls across all questions)
- Fans out **every question as an individual Haiku call** in parallel (up to 20 concurrent)
- Each question gets focused attention -- no degradation over long outputs
- Collects results into the standard block output format

This is **faster** (parallel tiny calls vs. one huge sequential call), **cheaper** (~$0.10/loan vs. ~$20/loan with Sonnet), and **more reliable** (each question evaluated in isolation).

Blocks that require complex reasoning or human judgment (EPD Review, Information Integrity) use **Sonnet** with the **plan-then-confirm** pattern instead.

Tools (System Queries)

30 tools map to API endpoints across the 9 lending systems. Agents use tools to retrieve loan data, credit reports, appraisals, and other documents needed to evaluate each question.

The Two-Phase Confirm Pattern

Blocks with `authority: confirm` (EPD Review, Information Integrity) use a plan-then-confirm flow:

- **Plan phase** -- Sonnet gathers data with read-only tools and identifies potential issues
- **Human review** -- Findings are presented in the Review Center escalation queue
- **Execute phase** -- Upon approval, the agent finalizes the deficiency findings

The 9 Lending Systems

All systems serve synthetic but realistic data from a **shared loan registry** ensuring data consistency -- the same borrower name, property address, and loan terms appear identically across all systems.

System	Port	Purpose
Loan Origination (LOS)	8101	Loan applications, borrower data, AUS findings, conditions
Document Vault	8102	Document lists, extracted data from income/asset/closing docs
Credit Service	8103	Tri-merge credit reports, liabilities, supplements
Appraisal Service	8104	Appraisal reports, comparable sales, CU scores
Employment Verification	8105	VVOE, tax returns, employment history
Title & Insurance	8106	Title commitments, insurance policies
Compliance Engine	8107	ATR/QM, TRID, HMDA, ECOA data
Investor Guidelines	8108	Fannie Mae/Freddie Mac/FHA/VA/RHS requirements
Flood Determination	8109	Flood zone determinations (SFHDF)

Event-Driven Orchestration

When a loan is submitted for QC review, the LOS publishes a `loan.submitted_for_qc` event to the `qc.events` Redis stream with the loan type, purpose, and phase. The gateway matches the event against route trigger conditions and launches the appropriate workflow.

```
LOS publishes "loan.submitted_for_qc" {loan_type: "Conventional", loan_purpose: "Purcha
  -> Gateway matches: conventional-purchase-pc route
  -> DAG executes: intake -> fan_out -> [16 blocks x ~50 Haiku calls each] -> merge ->
  -> ~845 questions evaluated across 9 systems in minutes
```

Demo: Planted Deficiencies

The mock systems include 8 loans with planted deficiencies across the portfolio of 50 loans:

- **Loan 2025-001** (Conventional): Income mismatch -- W-2 shows $95,000 but application states $98,500
- **Loan 2025-005** (Conventional): Missing flood insurance -- property in Zone AE (high-risk SFHA)

- **Loan 2025-008** (Conventional): VVOE dated 15 business days before note (exceeds 10-day requirement)
- **Loan 2025-012** (Conventional): DTI at 46.2% exceeds 45% guideline without compensating factors
- **Loan 2025-015** (VA): VA Certificate of Eligibility expired before closing
- **Loan 2025-020** (Conventional): Undisclosed $800/month car lease found on credit report
- **Loan 2025-025** (Conventional): Appraisal comparable 1.2 miles away (exceeds 1-mile suburban limit)
- **Loan 2025-030** (Conventional): Gift letter missing donor's statement of no repayment expectation

What This Demonstrates

- **AI as middleware** -- Agents coordinate across 9 independent lending systems, replacing the human "stare and compare" review process
- **Structured autonomy** -- Fixed QC workflows (YAML routes) with intelligent per-question analysis (Haiku) at each check
- **Human-in-the-loop** -- Plan-then-confirm for fraud/integrity findings; full autonomy for routine checks
- **Composability** -- 20 reusable blocks assembled into routes per loan type; new workflows created by recombining blocks
- **Real questions** -- 845 actual QC question codes from lender questionnaires, not synthetic checks
- **Cost efficiency** -- Per-question Haiku execution: ~$0.10/loan vs. ~$20/loan with monolithic Sonnet calls
- **Full visibility** -- Every tool call, LLM interaction, and finding is audited and visualized in the dashboard